

Q. Multi-Channel Decimation for Massive MIMO

Simulate multi-channel decimation in a Massive MIMO receiver and analyze computational reduction. (25) (BL4) (WK4)

Given Data:

- Number of Channels = 16
- Sampling Rate = 122.88 MHz
- Decimation Factor =

4 Tasks:

1. Generate multi-channel signals.
2. Design a common decimation filter.
3. Perform decimation.
4. Analyze computational reduction.
5. Compare channel spectra.
6. Discuss implementation benefits.

Aim:

1. To simulate 16-channel decimation in a Massive MIMO receiver.
2. To generate multi-channel signals at a 122.88 MHz sampling rate.
3. To design and apply a common FIR low-pass decimation filter.
4. To perform decimation by a factor of 4.
5. To analyze computational reduction and spectrum changes before and after decimation.

Objectives:

1. To generate 16 independent channel signals for a Massive MIMO receiver.
2. To design and apply a common 64th-order FIR low-pass filter.
3. To perform decimation by a factor of 4 on all channels.
4. To evaluate the reduction in sampling rate and computational load.
5. To compare the frequency spectra before and after decimation.

Tools Used: MATLAB

Code :

```
clc;
clear;
close
all;

%% Given Data
Fs = 122.88e6;    % Sampling Frequency
(Hz) D = 4;      % Decimation Factor
Nch = 16;        % Number of
Channels N = 4096; % Number of
Samples
t = (0:N-1)/Fs;

%% Generate Multi-Channel Signals
signals = zeros(Nch, N);

for k = 1:Nch
    f = 1e6 + (k-1)*0.5e6;    % Different frequency for each
    channel signals(k,:) = sin(2*pi*f*t) + 0.05*randn(1,N);
end

%% Design Common Decimation Filter
Fc = Fs/(2*D);          % Cutoff Frequency
h = fir1(64, Fc/(Fs/2)); % 64th-order FIR LPF

%% Perform Decimation
decSignals = zeros(Nch,
ceil(N/D));
```

```
for k = 1:Nch  
    temp = filter(h,1,signals(k,:));
```

```

    decSignals(k,:) =
temp(1:D:end); end

%% Computational
Reduction originalRate = Nch
* Fs; newRate = Nch *
(Fs/D);

reduction = (originalRate - newRate)/originalRate * 100;

fprintf('Original Sampling Rate : %.2f MHz\n', Fs/1e6);

fprintf('Output Sampling Rate : %.2f MHz\n', (Fs/D)/1e6);
fprintf('Computational Reduction: %.2f %%\n', reduction);

%% Compare Spectra (Channel 1)
figure;

subplot(2,1,1)
plot(abs(fft(signals(1,:))));
title('Spectrum Before
Decimation'); xlabel('Frequency
Bin'); ylabel('Magnitude');

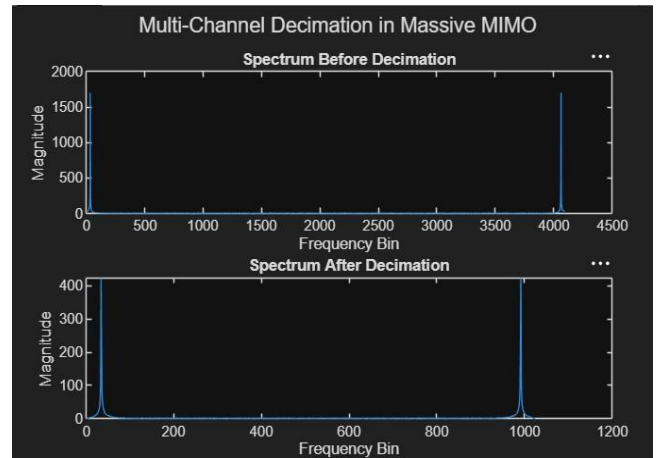
subplot(2,1,2)
plot(abs(fft(decSignals(1,:))));
title('Spectrum After
Decimation'); xlabel('Frequency
Bin'); ylabel('Magnitude');

sgtitle('Multi-Channel Decimation in Massive MIMO');

```

Output:

```
Original Sampling Rate : 122.88 MHz  
Output Sampling Rate  : 30.72 MHz  
Computational Reduction: 75.00 %  
>>
```



Conclusion:

The 16-channel Massive MIMO decimation system was successfully simulated using a common 64th-order FIR low-pass filter. The signals were decimated by a factor of 4, reducing the sampling rate from 122.88 MHz to 30.72 MHz. This resulted in a 75% reduction in sampling-rate-related computational load, while maintaining the required channel information. The spectra before and after decimation were compared to observe the effect of the process. Overall, the simulation demonstrates that multi-channel decimation can significantly reduce processing requirements and improve the efficiency of Massive MIMO receiver systems.